

COMPLETE PROJECT DOCUMENTATION

CONSTRUCTION, RENOVATION, REDEVELOPMENT, PROPERTY & NRI SERVICES PLATFORM

Project Type: Full-Stack Construction & Property Services Website + CMS + CRM + AI Assistant + Payment Platform + Project Management System

Architecture: Django-based Dynamic Web Application

Database: MySQL

Frontend: HTML5, CSS3, Bootstrap 5, Vanilla JavaScript

AI: Google AI Studio / Gemini

Primary Objective: Build a professional, scalable, secure and fully dynamic digital platform for construction, renovation, demolition, reconstruction, redevelopment, property, land, NRI and related services.

1. PROJECT OVERVIEW

The project is a complete digital platform for a construction and property-services organization.

It is not intended to function as a simple static company website.

The system combines:

- Professional public website
- Dynamic CMS
- Administrative dashboard
- CRM
- AI chatbot
- Lead management
- Customer management
- Service management
- Construction project management
- Site-visit management
- Estimate management
- Proposal management
- Payment gateway management

- UPI and QR payments
- Payment requests
- Payment receipts
- Blog and knowledge centre
- Project portfolio
- Document management
- Notifications
- Reports and analytics
- SEO management
- User and role management
- Audit logging
- Customer portal
- AI-assisted business operations

The complete system should operate as one connected platform.

2. PRIMARY BUSINESS OBJECTIVE

The platform must convert website visitors into qualified business opportunities and manage those opportunities through the complete customer lifecycle.

The overall business funnel is:

ATTRACT



UNDERSTAND



ENGAGE



QUALIFY



CAPTURE LEAD



FOLLOW UP



SITE VISIT



REQUIREMENT CONFIRMATION



ESTIMATE



PROPOSAL



NEGOTIATION



PAYMENT



PROJECT



MILESTONE MANAGEMENT



COMPLETION



CUSTOMER FEEDBACK



REPEAT BUSINESS / REFERRAL

The website, AI assistant, CRM and payment system must work together to support this funnel.

3. TECHNOLOGY REQUIREMENTS

The project must use only the following core technologies:

Python

Django

MySQL

HTML5

CSS3

Bootstrap 5

Vanilla JavaScript where necessary

Google AI Studio / Gemini for AI functionality

The following technologies must NOT be introduced:

React

Next.js

Node.js

PHP

Laravel

Vue

Angular

Tailwind CSS

MongoDB

Other frontend frameworks

The application should use Django's native architecture wherever practical.

4. SYSTEM ARCHITECTURE

The platform should be divided into logical Django applications.

Recommended structure:

core

users

pages

services

projects

enquiries

crm

customers

estimates

proposals

payments

blog

media

testimonials

faq

locations

notifications

documents

reports

cms

ai_assistant

seo

settings

audit

The existing project structure must be inspected before creating new applications.

Existing functionality and models should be reused whenever appropriate.

Duplicate models must not be unnecessarily created.

5. PUBLIC WEBSITE

The public website is the primary customer acquisition interface.

It must provide a premium, trustworthy and professional construction-industry experience.

The design must communicate:

Trust

Experience

Professionalism

Technical capability

Transparency

Quality

Reliability

Customer service

Project expertise

The website must be fully responsive across:

Desktop

Laptop

Tablet

Mobile

6. MAIN WEBSITE NAVIGATION

Recommended primary navigation:

Home

About Us

Construction

Renovation

Demolition & Reconstruction

Redevelopment

Property & Land

NRI Services

Projects

How We Work

Blog

Contact

A prominent CTA should be available:

Get a Free Consultation

Request Site Visit

Start Your Project

Talk to Our Team

The exact CTA can be configured through the CMS.

7. HOME PAGE

The homepage should act as the primary conversion page.

Recommended structure:

Hero Section

Business Value Proposition

Primary CTA

Secondary CTA

Service Categories

Construction Services

Renovation Services

Demolition & Reconstruction

Redevelopment

Property & Land

NRI Services

Documentation Assistance

Construction Finance Assistance

Why Choose Us

How We Work

Featured Projects

Before & After Projects

Service Locations

Customer Testimonials

Frequently Asked Questions

AI Assistant CTA

Blog / Knowledge Centre

Final Enquiry CTA

Footer

The homepage must be dynamic and CMS-controlled.

8. ABOUT US

The About page should communicate:

Company overview

Business history

Vision

Mission

Values

Experience

Capabilities

Construction approach

Quality standards

Customer commitment

Safety philosophy

Project management approach

Service locations

Team

Certifications where applicable

The CMS must allow administrators to edit this content without modifying source code.

9. CONSTRUCTION SERVICES

Construction must be a major service category.

Residential Construction

Independent House

New Home

G+1 Construction

Multi-Floor Residential

Custom Home Construction

Apartment Construction

New Apartment

Multi-Unit Residential

Apartment Building

Residential Complex

Villa Construction

Individual Villa

Premium Villa

Custom Villa

Luxury Villa

Commercial Construction

Office Buildings

Shops

Retail Spaces

Commercial Complexes

Business Buildings

Industrial / Structural Construction

Industrial Buildings

Structural Works

Specialized Construction

Building Extensions

Additional Floors where technically and legally applicable

Each service must have:

Service description

Benefits

Scope

Typical requirements

Process

FAQs

Related services

Gallery

Projects

Enquiry CTA

Site-visit CTA

Payment configuration where applicable

SEO settings

10. FLAT & APARTMENT RENOVATION

Renovation must be a dedicated major section rather than being treated as a minor construction service.

Flat Renovation

Complete Flat Renovation

Old Flat Renovation

Damaged Flat Renovation

Resale Flat Renovation

Rental Flat Renovation

Apartment Renovation

Complete Apartment Renovation

Individual Apartment Renovation

Multi-Apartment Renovation

Common Area Renovation

Interior Renovation

Kitchen

Bathroom

Flooring

False Ceiling

Painting

Electrical

Plumbing

Doors and Windows

Interior Modifications

Storage

Lighting

Other interior improvements

Structural Renovation

Structural Repairs

Wall Modifications

Slab Repairs

Water Leakage

Damaged Building Repairs

Strengthening

Restoration

Structural Assessment

The system must distinguish cosmetic renovation from potentially structural work.

The AI must never diagnose structural safety.

11. DEMOLITION & RECONSTRUCTION

Building Demolition

Residential

Commercial

Old Buildings

Damaged Buildings

Flat / Apartment Demolition

Internal Demolition

Partial Demolition

Complete Demolition

Damaged Property Demolition

Site Clearing

Debris Removal

Site Clearing

Construction Waste Management

Site Preparation

Reconstruction

Reconstruction after Demolition

Partial Reconstruction

Complete Reconstruction

Structural Reconstruction

The system should allow customers to enquire about demolition separately or as part of a complete demolition + reconstruction project.

12. REDEVELOPMENT

Redevelopment is a separate service category.

Property Redevelopment

Old Property

Damaged Property

Underutilized Property

Apartment / Building Redevelopment

Old Apartment

Residential Building

Multi-Unit Property

Demolition + Reconstruction

Assessment

Feasibility

Planning

Demolition

Site Preparation

New Construction

Project Supervision

Redevelopment enquiries should receive specialized CRM categorization.

13. PROPERTY & LAND SERVICES

Land Services

Buying Assistance

Selling Assistance

Land Inspection

Land Management

Property Services

Property Buying

Property Selling

Property Inspection

Property Maintenance

Property Management

The system must support property-service enquiries even when no construction work is immediately required.

14. NRI PROPERTY SERVICES

Dedicated NRI services must include:

Property Management

Land Management

Property Inspection

Construction Supervision

Renovation Supervision

Demolition Supervision

Reconstruction Supervision

Buying Assistance

Selling Assistance

Documentation Assistance

Government Coordination

NRI customers should have dedicated CRM categorization.

The system should allow:

NRI status

Country

Preferred communication method

Time-zone considerations

Remote supervision requirements

Authorized representative information where applicable

15. DOCUMENTATION & GOVERNMENT ASSISTANCE

This service category may include:

Property documentation assistance

Construction documentation

Government coordination

Application assistance

Approval coordination

Registration-related assistance where legally appropriate

Property records support

Other documentation services

The system must clearly state that the company does not provide unauthorized legal advice.

16. CONSTRUCTION LOAN & FINANCE ASSISTANCE

The platform should provide information and enquiry support for:

Construction loans

Property finance

Renovation finance

Project finance assistance

Loan documentation assistance

Finance consultation

The platform must not guarantee loan approval.

Financial advice must be presented carefully.

AI must not claim to approve loans or represent itself as a regulated financial advisor.

17. HOW WE WORK

Create a dedicated process page.

Recommended stages:

01 Consultation

02 Requirement Understanding

03 Property / Site Assessment

04 Feasibility

05 Planning

06 Estimate

07 Proposal

08 Customer Approval

09 Agreement

10 Advance Payment

11 Project Execution

12 Quality Monitoring

13 Milestone Updates

14 Final Inspection

15 Handover

16 Support

The stages must be CMS-editable.

18. PROJECTS / PORTFOLIO

Create a dynamic project portfolio.

Project categories:

Residential

Apartment

Villa

Commercial

Industrial

Renovation

Demolition

Reconstruction

Redevelopment

Property Services

NRI Projects

Each project should support:

Project name

Slug

Category

Project type

Location

Area

Description

Client type where permitted

Project status

Start date

Completion date

Scope

Challenges

Solutions

Materials

Before images

During-project images

After images

Gallery

Videos where supported

Testimonials

SEO metadata

Related services

19. BEFORE & AFTER PROJECTS

Renovation and redevelopment projects should support visual comparison.

Provide:

Before image

After image

Project description

Problem

Solution

Result

The public interface can include an interactive before/after slider using Bootstrap and Vanilla JavaScript.

20. BLOG / KNOWLEDGE CENTRE

Create a complete dynamic Blog CMS.

Blog functionality:

Blog categories

Tags

Authors

Featured image

Content editor

SEO title

Meta description

Slug

Excerpt

Publishing date

Updated date

Draft

Published

Scheduled publishing where implemented

Featured article

Related articles

Search

Pagination

Category filtering

Tag filtering

Social sharing

Related services

The Blog should support construction and property knowledge content.

Possible categories:

Construction

Home Building

Renovation

Apartment Renovation

Demolition

Reconstruction

Redevelopment

Property

Land

NRI Property

Construction Finance

Government Documentation

Home Maintenance

Architecture

Interior

Project Management

21. FAQ MANAGEMENT

Create a dynamic FAQ CMS.

Administrators can create:

Question

Answer

Category

Service

Display order

Published status

Featured status

FAQs can appear on:

Homepage

Service pages

Blog pages

Contact page

Dedicated FAQ page

AI knowledge base

22. TESTIMONIAL MANAGEMENT

Create:

Testimonials

Customer name

Location

Service

Project

Testimonial

Photo where permitted

Rating

Featured

Published

Display order

Testimonials can be displayed dynamically throughout the website.

23. ENQUIRY MANAGEMENT

Every major CTA must connect to the enquiry system.

Enquiry sources:

Website

Contact form

Service page

Project page

Blog

AI chatbot

WhatsApp

Phone

Manual CRM entry

Campaign

Referral

Other

Capture:

Name

Phone

Email

WhatsApp

Location

Service

Property type

Property condition

Project description

Budget

Timeline

Preferred contact method

NRI status

Source

Message

Attachments where applicable

24. CRM

The CRM is the central business-management component.

The CRM should manage:

Leads

Customers

Enquiries

Follow-ups

Site visits

Estimates

Proposals

Payments

Projects

Documents

Notes

Tasks

Activities

Communication history

The CRM must provide a complete customer lifecycle.

25. CRM LEAD WORKFLOW

Recommended status:

NEW

CONTACTED

QUALIFIED

SITE VISIT

REQUIREMENT CONFIRMED

ESTIMATE PREPARED

PROPOSAL SENT

NEGOTIATION

WON

LOST

ON HOLD

The workflow must be configurable.

26. LEAD MANAGEMENT

Each lead should contain:

Lead ID

Customer

Service

Property type

Location

Requirement

Budget

Timeline

Lead source

Priority

Assigned staff

Status

Next follow-up

Notes

Documents

AI summary

Created date

Updated date

Activity history

27. CUSTOMER MANAGEMENT

Create a customer profile containing:

Name

Phone

Email

WhatsApp

Address

City

State

Country

NRI status

Preferred contact

Projects

Leads

Enquiries

Payments

Documents

Communication history

Site visits

Estimates

Proposals

Notes

28. LEAD PRIORITY

Support:

LOW

MEDIUM

HIGH

URGENT

AI may suggest priority based on:

Timeline

Project value indicators

Requirement completeness

Customer engagement

Site-visit request

Urgency

Staff can override AI priority.

29. FOLLOW-UP MANAGEMENT

CRM staff must be able to create:

Follow-up date

Follow-up time

Assigned staff

Follow-up type

Notes

Outcome

Next action

Reminder

Status

Follow-ups should appear on the dashboard.

30. SITE VISIT MANAGEMENT

Site visits should support:

Customer

Lead

Project

Property

Location

Date

Time

Assigned staff

Site engineer / representative

Status

Notes

Attachments

Site photographs

Assessment

Next action

Site visit fee

Payment status

Statuses:

REQUESTED

SCHEDULED

CONFIRMED

COMPLETED

RESCHEDULED

CANCELLED

31. ESTIMATE MANAGEMENT

CRM users must be able to create estimates.

Estimate fields:

Estimate number

Customer

Lead

Project

Service

Items

Quantity

Unit

Unit price

Subtotal

Tax

Discount

Total

Validity

Terms

Notes

Attachments

Status

Estimate versions

The customer should receive a professional estimate.

32. PROPOSAL MANAGEMENT

Proposal functionality:

Proposal number

Customer

Lead

Project

Scope

Deliverables

Timeline

Commercial terms

Payment schedule

Exclusions

Terms

Validity

Attachments

Approval status

Customer response

Version control

Statuses:

DRAFT

SENT

VIEWED

ACCEPTED

REJECTED

EXPIRED

NEGOTIATION

33. CUSTOMER APPROVAL

Where appropriate, customers should be able to:

View proposal

View estimate

Accept

Reject

Request changes

Download PDF

Proceed to payment

The system should record approval activity.

34. PAYMENT SYSTEM

Create a gateway-agnostic payment module.

It must support:

Third-party gateways

Hosted checkout

Embedded checkout

HTML payment integration where supported

JavaScript checkout where supported

UPI

UPI QR

Payment links

Bank transfer

Manual payment
Consultation payment
Site-visit payment
Advance payment
Milestone payment
Final payment
Refund tracking

35. PAYMENT GATEWAY CONFIGURATION

Administrators should be able to add gateways through the dashboard.

Gateway configuration may include:

Gateway name
Provider
Gateway type
Public key
Merchant ID
Checkout URL
API credentials
Webhook configuration
Success URL
Failure URL
Cancel URL
Documentation URL
Test mode
Live mode
Active status

For providers supporting safe client-side integration, administrators can configure approved HTML/embed code.

For gateways requiring server-side API authentication or signature verification, a Django gateway adapter must be implemented.

The system must never expose secret credentials in frontend code.

36. UPI PAYMENT

UPI configuration:

UPI ID

Merchant name

QR code

Payment instructions

Active status

The system should support:

UPI payment URI

QR code

Mobile UPI flow

Manual UPI verification

Gateway-based UPI where supported

A customer clicking a UPI button must not automatically be considered as having paid.

Payment verification must happen through the backend/gateway or authorized manual verification.

37. SERVICE PAYMENT CONFIGURATION

Every service may have its own payment configuration.

Examples:

Consultation fee

Site-visit fee

Booking amount

Advance percentage

Project milestone

Custom payment amount

Payment enabled/disabled

Available payment methods

Payment gateway

Minimum amount

Maximum amount

The CMS administrator controls these settings.

38. CRM PAYMENT REQUESTS

CRM staff can generate payment requests.

Payment request includes:

Customer

Lead

Project

Service

Estimate

Proposal

Purpose

Amount

Tax

Discount

Total

Due date

Payment method

Gateway

Notes

A secure payment link is generated.

39. PROJECT MILESTONE PAYMENTS

Construction projects can contain configurable milestones.

Example:

Booking Advance

Foundation

Structure

Brickwork

MEP

Finishing

Handover

The percentage and amount must be configurable per project.

40. PAYMENT TRANSACTIONS

Every payment should have:

UUID

Payment reference

Customer

Lead

Project

Service

Gateway

Payment method

Amount

Tax

Discount

Currency

Status

Provider transaction ID

Provider order ID

Failure reason

Created date

Completed date

41. PAYMENT STATUS

Support:

CREATED

INITIATED

PENDING

AUTHORIZED

SUCCESS

FAILED

CANCELLED

REFUNDED

PARTIALLY REFUNDED

EXPIRED

MANUAL REVIEW

42. PAYMENT VERIFICATION

Never trust the frontend.

Successful payment must be verified through:

Gateway API

Webhook

Server-side verification

Authorized manual verification

Implement:

Webhook signature verification

Idempotency

Duplicate transaction protection

Server-side amount validation

Secure transaction references

Audit logging

43. PAYMENT RECEIPTS

Generate professional receipts containing:

Company details

Customer

Service

Project

Payment reference

Transaction ID

Amount

Tax

Discount

Total paid

Payment method

Payment date

Status

Receipt number

The system should provide:

View receipt

Print receipt

PDF receipt

44. REFUND MANAGEMENT

Support:

Refund request

Refund approval

Partial refund

Full refund

Gateway refund

Manual refund tracking

Refund reason

Refund reference

Refund status

Refund audit history

45. CUSTOMER PORTAL

Add a secure customer portal.

Customers can view:

Profile

Enquiries

Leads

Site visits

Estimates

Proposals

Projects

Project updates

Payments

Receipts

Documents

Messages

Notifications

Support requests

The customer should only access records belonging to their account.

46. PROJECT MANAGEMENT

Once a lead becomes a confirmed project, create a project record.

Project information:

Project ID

Customer

Service

Property

Location

Project type

Start date

Expected completion

Actual completion

Project manager

Team

Budget

Contract amount

Status

Milestones

Tasks

Documents

Payments

Progress

Photos

Reports

47. PROJECT STATUS

Possible statuses:

PLANNING

APPROVED

READY TO START

IN PROGRESS

ON HOLD

DELAYED

NEAR COMPLETION

COMPLETED

CANCELLED

The status workflow must be configurable.

48. PROJECT TASK MANAGEMENT

Tasks may include:

Site preparation

Demolition

Foundation

Structural work

Brickwork

Electrical

Plumbing

MEP

Flooring

Painting

Interior

Inspection

Final finishing

Handover

Each task can have:

Assigned person

Start date

Due date

Status

Priority

Notes

Attachments

49. PROJECT PROGRESS

Track:

Overall percentage

Milestone completion

Task completion

Site photographs

Progress notes

Delays

Issues

Approvals

Customer updates

50. DOCUMENT MANAGEMENT

Create centralized document management.

Documents may include:

Estimates

Proposals

Contracts

Invoices

Receipts

Property documents

Site reports

Approval documents

Payment proofs

Project documents

Customer-uploaded files

Admin-uploaded files

Each document should have:

Document type

Owner

Related lead

Related customer

Related project

Upload date

Version

Visibility

Permissions

51. MEDIA LIBRARY

Create centralized media management.

Support:

Images

Project photos

Before/after images

Service images

Blog images

Team images

Documents

PDF files

Media metadata

Alt text

Caption

Title

Category

Upload date

The CMS should allow media reuse across pages.

52. CMS

The CMS must control the public website dynamically.

Administrators should be able to manage:

Pages

Sections

Services

Categories

Projects

Testimonials

FAQs

Blog

Menus

Footer

Contact details

Business information

SEO

Media

Service locations

Payment settings

AI knowledge

Website settings

CTA buttons

Forms

53. DYNAMIC PAGE MANAGEMENT

Create a flexible page model.

Each page should support:

Title

Slug

Content

Featured image

Sections

CTA

SEO title

Meta description

Keywords

Canonical URL

Open Graph image

Published status

Publish date

Template

Order

The goal is to minimize developer dependency for content changes.

54. SERVICE CMS

Each service should be fully editable.

Fields:

Service name

Slug

Category

Short description

Full description

Benefits

Scope

Process
Requirements
FAQs
Images
Projects
Related services
CTA
Payment settings
SEO
Published status
Display order

55. SERVICE AREA MANAGEMENT

Create dynamic service locations.

Support:

Country

State

City

Area

PIN code where appropriate

Service availability

Location description

SEO metadata

Examples may include Bangalore and other cities/regions served by the business.

Do not hard-code locations.

56. SEO MANAGEMENT

Every major public page must support:

SEO title

Meta description

Canonical URL

Open Graph title

Open Graph description

Open Graph image

Schema configuration where appropriate

Slug

Robots settings

Image alt text

The CMS must allow SEO content to be edited.

57. SEO-FRIENDLY URLS

Use clean URLs.

Examples:

/construction/

/construction/residential/

/construction/villa/

/renovation/

/renovation/flat-renovation/

/demolition-reconstruction/

/redevelopment/

/property-land/

/nri-services/

/projects/

/blog/

/contact/

Service and project slugs must be dynamic.

58. STRUCTURED DATA / SCHEMA

Where appropriate, support:

Organization

LocalBusiness

Service

FAQPage

Article

BreadcrumbList

WebSite

Review

Project-related structured data where appropriate

Schema must contain accurate information only.

59. AI CONSTRUCTION & PROPERTY ASSISTANT

The public AI chatbot should function as:

AI Construction & Property Consultant

It should help visitors understand:

Construction

Renovation

Demolition

Reconstruction

Redevelopment

Property

Land

NRI services

Documentation

Construction finance

Site visits

Payments

Project process

The AI should guide customers toward the correct next step.

60. AI CHATBOT QUICK ACTIONS

Recommended options:

Build a New Property

Renovate a Flat / Apartment

Demolish a Building
Reconstruct a Property
Redevelop a Property
Property / Land Services
NRI Property Assistance
Construction Finance
Request Site Visit
Make a Payment
Talk to Our Team

61. AI LEAD QUALIFICATION

The AI should gradually collect:

Name

Phone

Email

WhatsApp

Location

Property type

Service

Property area

Project description

Budget range

Timeline

Property condition

NRI status

Preferred contact

Site visit requirement

Do not ask for everything in one message.

62. PROPERTY CONDITION

Options:

Good but needs updating

Old

Damaged

Partially damaged

Severely damaged

Partially demolished

Already demolished

Not sure

AI must not diagnose structural safety.

63. BUDGET RANGE

Configurable options such as:

Under ₹5 lakh

₹5–10 lakh

₹10–25 lakh

₹25–50 lakh

₹50 lakh–₹1 crore

₹1 crore+

Not sure

These values must be CMS/configuration controlled.

64. PROJECT TIMELINE

Options:

Immediately

Within 1 month

1–3 months

3–6 months

6–12 months

Researching

65. AI LEAD CREATION

Once sufficient information is collected:

Create CRM Lead

Attach customer information

Attach service

Attach property details

Attach budget

Attach timeline

Attach location

Attach AI intent

Generate AI summary

Set suggested priority

Save conversation reference

Store source as:

AI CHATBOT

66. AI CONVERSATION SUMMARY

Example internal summary:

Customer is planning a complete renovation of a 3BHK apartment. Property is approximately 15 years old and requires kitchen, bathroom, flooring and electrical upgrades. Customer is located in Bangalore and wants the project to begin within two months. Site visit requested.

The summary must be visible inside CRM.

67. AI HUMAN HANDOFF

AI must hand over to a human when:

Customer requests a person

Customer is frustrated

AI lacks information

Complex project requirement

Pricing requires assessment

Structural safety issue

Legal issue

Property dispute

Loan approval question

Other sensitive situations

Show:

HUMAN AGENT ACTIVE

During takeover, AI must stop responding.

Staff can return the conversation to AI.

68. AI KNOWLEDGE BASE

AI should use approved CMS information.

Knowledge categories:

Services

Service descriptions

Process

FAQs

Business information

Locations

Contact details

Policies

Payment information

General construction information

The AI must not invent company services, prices, guarantees or policies.

69. AI SAFETY

AI must never:

Claim to be human

Guarantee structural safety

Provide unauthorized legal advice

Guarantee government approval

Guarantee loan approval

Invent construction prices

Confirm unverified payments

Approve refunds

Modify CRM payment status without authorization

Expose private information

Expose API keys

Expose customer information to other customers

70. AI PAYMENT ASSISTANCE

AI should understand:

I want to pay

How can I pay?

Can I pay by UPI?

Where is the QR code?

I need my receipt

My payment is pending

My payment failed

I want to pay the next milestone

AI must retrieve actual payment status from the Django backend.

71. AI ADMIN DASHBOARD

Dashboard metrics:

AI conversations

Unique visitors

New AI leads

Qualified leads

Site visit requests

Human handoffs

Won leads

Estimated pipeline

Payment-related conversations

Top service intents

Top locations

Common questions

Abandoned conversations

72. AI CONVERSATION MANAGEMENT

Admin can view:

Customer

Intent

Service

Location

Lead

Conversation status

AI summary

Collected information

Transcript

CRM activities

Follow-ups

Site visits

Payment status

Staff assignment

73. ADMIN DASHBOARD

The main admin dashboard should provide a business overview.

Cards:

Total Leads

New Leads

Qualified Leads

Site Visits

Open Proposals

Active Projects

Pending Payments

Revenue

AI Leads

Conversion Rate

Follow-ups Today

74. ADMIN SIDEBAR

Recommended:

Dashboard

AI Conversations

Leads

Customers

Follow-ups

Site Visits

Estimates

Proposals

Payments

Projects

Services

Pages

Blog

FAQs

Testimonials

Media Library

Documents

Locations

Reports

AI Settings

SEO

Notifications

Users & Permissions

Website Settings

Audit Logs

75. REPORTING & ANALYTICS

Reports should include:

Lead reports

Service demand

Lead source

Location demand

Conversion rate

Sales pipeline

Site visits

Estimate conversion

Proposal conversion

Project revenue

Payment revenue

Payment failures

Refunds

AI performance

Blog performance where analytics are available

76. LEAD SOURCE ANALYTICS

Sources:

Website

AI Chatbot

Google

Meta

WhatsApp

Phone

Referral

Campaign

Blog

Direct

Manual

Other

Show:

Leads

Qualified leads

Site visits

Proposals

Won

Revenue

Conversion rate

77. SERVICE DEMAND ANALYTICS

Track demand for:

Construction

Renovation

Demolition

Reconstruction

Redevelopment

Property

Land

NRI

Documentation

Finance

Consultation

Show:

Enquiries

Leads

Qualified

Site visits

Proposals

Won

Revenue

78. NOTIFICATION SYSTEM

Notifications should support:

New enquiry

New AI lead

High-priority lead

Follow-up reminder

Site visit reminder

New payment

Failed payment

Payment pending

Proposal viewed

Proposal accepted

Refund request

New customer message

New document

Project milestone

The notification system should be extensible.

79. EMAIL / WHATSAPP / SMS READINESS

The architecture should allow integration with external communication providers.

Do not hard-code a specific provider.

Create an abstraction for:

Email

WhatsApp

SMS

The system can later integrate approved providers.

80. CONTACT MANAGEMENT

The website must display configurable:

Phone

WhatsApp

Email

Office address

Business hours

Service locations

Social media

Map information

The CMS administrator should be able to update these details.

81. ENQUIRY FORM BUILDER

Create a configurable form system where appropriate.

Admin can define:

Field

Label

Type

Required

Placeholder

Options

Display order

Service association

Active status

Potential field types:

Text

Phone

Email

Dropdown

Radio

Checkbox

Textarea

File upload

Date

Budget

Location

82. FORM SECURITY

All public forms must implement:

CSRF protection

Server-side validation

Input sanitization

Spam protection

Rate limiting where appropriate

File validation

Secure upload handling

Email validation

Phone validation

Never trust browser-side validation alone.

83. FILE UPLOAD SECURITY

Uploaded files must be validated for:

Extension

MIME type

File size

Filename

Storage location

Potential malicious content

Never allow uploaded files to execute as server-side code.

84. USER MANAGEMENT

Create role-based access.

Roles may include:

Super Administrator

Administrator

CRM Manager

Sales Manager

Sales Executive

Project Manager

Site Engineer

Finance Manager

Content Manager

Blog Manager

Payment Manager

Support Staff

AI Manager

Customer

Permissions must be configurable.

85. ROLE-BASED ACCESS CONTROL

Users should only see functionality they are authorized to access.

Examples:

Sales staff:

Leads

Customers

Follow-ups

Site visits

Estimates

Proposals

Payment requests

Project staff:

Projects

Tasks

Site reports

Documents

Progress

Finance staff:

Payments

Receipts

Refunds

Reports

Content staff:

Pages

Services

Blog

FAQs

Media

86. AUDIT LOGGING

Important system actions must be recorded.

Examples:

Login

Logout

Create

Edit

Delete

Publish

Unpublish

Payment verification

Refund approval

Gateway modification

UPI modification

Lead reassignment

Proposal approval

Project status change

User permission change

Store:

User

Action

Record

Timestamp

IP where appropriate

Before/after values where appropriate

87. SOFT DELETE

Critical business records should preferably use soft deletion where appropriate.

Examples:

Leads

Customers

Projects

Payments

Documents

Blog posts

Services

Instead of permanently deleting important records, mark them as inactive/deleted.

Payment records must never be casually deleted.

88. SEARCH SYSTEM

Global admin search should support:

Customers

Leads

Projects

Payments

Services

Blog

Documents

Proposals

Estimates

Search by:

Name

Phone

Email

Reference

ID

Service

Location

Status

89. FILTERING & PAGINATION

Admin tables must support:

Search

Filters

Sorting

Pagination

Date ranges

Status

Assigned staff

Service

Location

Bulk actions where safe

90. BULK ACTIONS

Safe bulk actions may include:

Assign leads

Change status

Publish blog posts

Unpublish content

Archive records

Export data

Bulk notification

Bulk category assignment

Do not allow unsafe bulk financial operations.

91. WEBSITE SEARCH

Public website search should support:

Services

Projects

Blog

FAQs

Knowledge content

Search results should be SEO-friendly and user-friendly.

92. CUSTOMER COMMUNICATION HISTORY

Store relevant communication records:

Phone follow-up notes

Email records where integrated

WhatsApp records where integrated

AI conversations

CRM notes

Site visit notes

Proposal communication

Payment communication

The system should provide a chronological timeline.

93. CRM ACTIVITY TIMELINE

For every lead/customer:

Enquiry

AI conversation

Call

Follow-up
Site visit
Estimate
Proposal
Payment
Project creation
Project milestone
Document
Message
Status change
Display chronologically.

94. PROJECT CUSTOMER UPDATES

Customers should be able to receive project updates such as:

Milestone completed
Site progress
Upcoming work
Payment due
Payment received
Inspection scheduled
Document uploaded
Project completed

95. CUSTOMER SUPPORT

Add a basic support/contact mechanism.

Customer can submit:

Question
Issue
Project concern
Payment issue
Document request

General support

Each support request should receive a ticket/reference number.

96. SUPPORT TICKETS

Ticket fields:

Ticket ID

Customer

Project

Category

Priority

Subject

Description

Attachments

Assigned staff

Status

Created date

Updated date

Statuses:

OPEN

IN PROGRESS

WAITING FOR CUSTOMER

RESOLVED

CLOSED

97. BLOG SEO

Blog posts must support:

SEO title

Meta description

Slug

Canonical URL

Featured image

Alt text

Article schema

Open Graph metadata

Related services

Internal linking

Author

Publish date

Updated date

98. CONTENT WORKFLOW

Content statuses:

DRAFT

REVIEW

APPROVED

PUBLISHED

ARCHIVED

Users should only publish content if they have permission.

99. SCHEDULING

Where appropriate, support scheduled publication for:

Blog posts

Pages

Projects

Announcements

Do not require scheduled publishing for every object unless necessary.

100. LEGAL PAGES

Create editable legal pages:

Privacy Policy

Terms & Conditions

Cookie Policy where applicable

Payment Terms

Refund Policy

Cancellation Policy

Disclaimer

The exact legal language should be reviewed by the organization's legal advisor before production use.

101. COOKIE / PRIVACY READINESS

The website should provide appropriate privacy controls where required.

Do not collect unnecessary personal information.

Only collect data necessary for the relevant business function.

102. BACKUP & RECOVERY

Production deployment should include:

Database backups

Media backups

Configuration backup

Recovery procedure

Backup retention policy

The application must not rely on a single copy of business data.

103. SYSTEM SETTINGS

Create centralized settings.

Business:

Company name

Logo

Phone

Email

Address

Working hours

GST details where applicable

Social media

Contact settings

Website:

Theme

Header

Footer

Navigation

CTA

Favicon

SEO defaults

Payment:

Currency

Gateway

UPI

Tax

Receipt

AI:

AI name

Welcome message

Business description

Working hours

Human handoff

Knowledge base

Notifications

Email:

SMTP configuration

Templates

104. THEME MANAGEMENT

The frontend should maintain a consistent professional visual identity.

Recommended style direction:

Midnight navy

Warm white

Neutral gray

Subtle accent color

Professional typography

Clear visual hierarchy

Avoid excessive animation.

The website should look like a premium construction/property organization rather than a generic template.

105. LIGHT / DARK MODE

Where appropriate, provide:

Light mode

Dark mode

Theme preference

The implementation must preserve readability and accessibility.

106. RESPONSIVE DESIGN

Every page must work correctly on:

1920px desktop

1440px desktop

Laptop

Tablet

Mobile

The AI chatbot and payment pages require special mobile optimization.

107. ACCESSIBILITY

Follow practical accessibility standards.

Include:

Readable typography

Sufficient contrast

Keyboard navigation

Visible focus states

Alt text

Accessible labels

Form error messages

Semantic HTML

Proper heading hierarchy

Accessible buttons

108. PERFORMANCE

Optimize:

Images

CSS

JavaScript

Database queries

Django templates

Caching where appropriate

Pagination

Lazy loading

Media delivery

Avoid unnecessary JavaScript libraries.

109. SECURITY

Implement:

Django security settings

CSRF protection

XSS prevention

SQL injection prevention

Secure authentication

Password hashing

Permission controls

Session security
Secure cookies
HTTPS in production
File upload security
Rate limiting where appropriate
Webhook verification
Payment verification
Audit logging
Sensitive-data protection

110. DATABASE SECURITY

Use MySQL properly.
Use:
Indexes
Foreign keys
Constraints
Transactions
Appropriate field types
Unique constraints
Database-level integrity where appropriate
Avoid storing unnecessary sensitive information.

111. APPLICATION LOGGING

Create structured application logs for:
Errors
Payment events
Webhook events
Authentication
Critical admin actions
API failures
External service failures

Background operations

Do not log secrets or sensitive payment credentials.

112. ERROR HANDLING

Create professional error pages:

404

403

500

Payment failure

Payment pending

Permission denied

Session expired

Form validation errors

Do not expose technical stack traces to public users.

113. API-READY ARCHITECTURE

Although the initial frontend uses Django templates, the architecture should be designed so future integrations can expose controlled APIs.

Potential future integrations:

Mobile application

External CRM

Marketing systems

WhatsApp provider

Payment provider

Analytics

Partner systems

Do not introduce unnecessary API frameworks unless required.

114. AI-READY ARCHITECTURE

AI functionality should be isolated from core business logic.

AI should request controlled information from Django services.

The AI layer must not directly manipulate the database without business rules.

Example:

AI



Django AI Service



Business Logic



Database

This prevents uncontrolled AI actions.

115. AI ACTION PERMISSIONS

AI can:

Read approved service information

Answer FAQs

Collect enquiry information

Create lead through controlled service

Create conversation summary

Suggest priority

Guide to payment page

Read payment status

Suggest human handoff

AI cannot directly:

Change payment amount

Approve refunds

Delete customers

Change financial records

Grant permissions

Publish sensitive content

Change system configuration

116. ANALYTICS

Track appropriate business events:

Page views

Service views

Project views

Blog views

Enquiry submissions

AI conversations

AI leads

Site visit requests

Payment initiation

Payment success

Proposal views

Proposal acceptance

Project conversion

Use privacy-conscious analytics practices.

117. CONVERSION TRACKING

Track the complete funnel:

Visitor

Service view

CTA click

Enquiry

AI conversation

Lead

Qualified lead

Site visit

Estimate

Proposal

Payment

Project

Completion

This enables management to identify where leads are being lost.

118. MARKETING INTEGRATION READINESS

The system should be prepared for:

Google Ads

Meta Ads

WhatsApp campaigns

Email campaigns

SEO

Organic search

Referral campaigns

Campaign landing pages

UTM tracking

Lead source tracking

Do not hard-code marketing platforms into the core CRM.

119. CAMPAIGN TRACKING

Capture:

Campaign

Source

Medium

Term

Content

Landing page

First-touch source

Last-touch source

Where technically and legally appropriate.

120. WHATSAPP CTA

Every important conversion page should optionally include:

WhatsApp

Call

Enquiry

AI Chat

Site Visit

Payment

CTA buttons.

The destination numbers should be CMS-configurable.

121. DYNAMIC CTA SYSTEM

Administrators should be able to configure:

CTA label

URL/action

Service

Position

Icon

Visibility

Priority

The same CTA system can be reused throughout the website.

122. FOOTER

Footer should contain:

Company information

Services

Important links

Projects

Blog

Contact

WhatsApp

Privacy Policy

Terms

Refund Policy

Disclaimer

Social media

Copyright

123. ADMIN WEBSITE PREVIEW

CMS users should ideally be able to preview unpublished content before publishing.

Preview should not expose drafts publicly without secure authorization.

124. CONTENT VERSIONING

For important content, support version history where practical.

This is especially useful for:

Services

Pages

Blog

Payment terms

Policies

Proposals

125. SYSTEM HEALTH DASHBOARD

Administrator dashboard may include:

Database status

Application status

Media storage

Failed jobs if applicable

Webhook failures

Payment gateway status

AI service status

Email status

Recent errors

This helps identify operational problems early.

126. PAYMENT GATEWAY HEALTH

Display:

Gateway active/inactive

Test/live mode

Last successful transaction

Last webhook

Webhook failure count

Configuration status

Do not expose secret credentials.

127. AI SERVICE HEALTH

Display:

AI enabled

Model configuration

Last successful interaction

Failure count

Knowledge base status

Human handoff availability

Do not expose API credentials.

128. ADMIN HOME DASHBOARD BUSINESS VIEW

The dashboard should answer:

How many new leads arrived?

Which services are generating demand?

Which leads need follow-up?

How many site visits are scheduled?

How many proposals are pending?

How much payment is pending?

How much revenue was collected?

Which projects are active?

What is the AI generating?

Which enquiries need immediate attention?

129. STAFF WORK QUEUE

Create personalized work queues:

My New Leads

My Follow-ups Today

My Site Visits

My Proposals

My Payment Requests

High Priority Leads

Overdue Tasks

Project Issues

130. DASHBOARD QUICK ACTIONS

Provide:

Add Lead

Create Customer

Schedule Site Visit

Create Estimate

Create Proposal

Create Payment Request

Add Project

Add Service

Create Blog

Add FAQ

Upload Media

131. DATA EXPORT

Authorized administrators should be able to export:

Leads

Customers

Payments

Projects

Site visits

Estimates

Proposals

Blog

Reports

Exports should respect user permissions.

132. IMPORT

Where business requirements justify it, support controlled imports for:

Customers

Leads

Services

Projects

The import process must validate data before committing.

133. DUPLICATE DETECTION

Detect potential duplicate leads using combinations such as:

Phone

Email

Customer

Recent enquiry

The system should warn staff rather than automatically deleting records.

134. CUSTOMER CONSENT

Where personal data is collected, capture appropriate consent where required.

Maintain consent information according to the organization's privacy requirements.

135. PUBLIC USER REGISTRATION

Customer registration should be optional unless a portal feature requires it.

Users should not be forced to create accounts merely to submit a simple enquiry.

136. SECURE CUSTOMER PORTAL ACCESS

Use:

Django authentication

Password hashing

Secure sessions

Password reset

Email verification where appropriate

Permission checks

Object-level access control

Customers must never access another customer's data.

137. ADMIN LOGIN SECURITY

Implement:

Secure authentication

Strong passwords

Session security

Permission controls

Optional two-factor authentication readiness

Login auditing

Account lockout/rate limiting where appropriate

138. PROJECT DOCUMENT VISIBILITY

Documents should support visibility:

Admin Only

Staff

Customer

Specific Project Team

Specific Customer

Public where appropriate

Never expose private documents through predictable URLs.

139. PUBLIC PROJECT PRIVACY

Project information must respect customer privacy.

Do not publish:

Private phone numbers

Private addresses

Confidential documents

Personal identification information

Financial information

without authorization.

140. IMAGE MANAGEMENT

Project images should support:

Original

Optimized version

Thumbnail

Alt text

Caption

Credit

Before/after designation

SEO metadata

141. BLOG CONTENT EDITOR

The CMS editor should support:

Headings

Paragraphs

Lists

Links

Images

Tables where required

Quotes

Callouts

Internal links

External links

SEO fields

The system must sanitize submitted HTML appropriately.

142. BLOG RELATED SERVICES

Each blog article can be linked to one or more services.

Example:

Article:

How to Renovate an Old Apartment

Related services:

Flat Renovation

Apartment Renovation

Structural Repair

This improves user navigation and conversion.

143. INTERNAL LINKING

The CMS should make it easy to link:

Blog → Service

Service → Project

Project → Service

FAQ → Service

Service → Contact

Service → AI Assistant

Service → Payment

This creates a connected content ecosystem.

144. AI KNOWLEDGE SYNCHRONIZATION

When approved CMS content changes, the AI knowledge layer should use the latest approved information.

Do not require developers to manually rewrite AI prompts every time a service description changes.

145. AI RESPONSE SOURCE CONTROL

AI answers should prioritize:

Approved CMS content

Approved FAQ

Approved service information

Approved business settings

Approved payment configuration

The AI should explicitly say when information requires human confirmation.

146. MULTILINGUAL READINESS

Initial language:

English

Architecture should be prepared for:

Kannada

Telugu

Hindi

Additional languages later.

The database/content structure should avoid creating unnecessary duplicate logic for every language.

147. INTERNATIONAL / NRI READINESS

The platform should support NRI customers while retaining INR as the primary business currency if required.

Potential future capability:

International payment gateways

Foreign currencies

International phone numbers

Country-specific service information

Timezone-aware communication

Do not enable these features unless configured.

148. PAYMENT CURRENCY

Default:

INR

Currency should be configurable.

Do not assume every gateway supports every currency.

Gateway capability must be respected.

149. FINANCE DATA SEPARATION

Payment data should be separated logically from general CRM notes.

Financial records must have stricter permissions.

150. PROJECT FINANCIAL SUMMARY

Authorized staff should see:

Contract amount

Paid amount

Pending amount

Milestone amount

Refund amount

Outstanding amount

Payment history

Do not expose financial data to unauthorized users.

151. PROJECT PROFITABILITY READINESS

The architecture may later support:

Estimated cost

Actual cost

Revenue

Margin

Vendor expenses

Material expenses

Labour expenses

Do not implement advanced accounting unless required, but keep the project model extensible.

152. VENDOR MANAGEMENT READINESS

Future module:

Vendors

Contractors

Suppliers

Materials

Purchase orders

Vendor payments

Vendor documents

This can be added without redesigning the core CRM.

153. INVENTORY READINESS

Future construction management can support:

Materials

Quantity

Unit

Supplier

Purchase

Usage

Stock

This is optional for the first release but should not conflict with the project architecture.

154. STAFF / TEAM MANAGEMENT

Team records may include:

Name

Designation

Department

Photo

Contact

Skills

Assigned projects

Active status

Public visibility

155. PROJECT TEAM ASSIGNMENT

Assign:

Project manager

Engineer

Supervisor

Sales executive

Support staff

Other authorized users

156. SERVICE TEAM ASSIGNMENT

Services may optionally have:

Default manager

Default team

Default CRM owner

Default payment method

Default notification group

157. AUTOMATION ENGINE

Create configurable business automations where useful.

Examples:

New enquiry → create lead

AI-qualified lead → notify sales

Site visit requested → notify assigned staff

Payment successful → update payment status

Proposal accepted → notify project team

Follow-up overdue → dashboard alert

Payment pending → customer notification

Do not build uncontrolled automation.

Every automation must have clear business rules.

158. SCHEDULED TASKS

Where required, support scheduled operations such as:

Follow-up reminders

Scheduled blog publishing

Payment expiry

Notifications

Reports

System maintenance

Use a suitable Django-compatible mechanism without introducing unnecessary frameworks.

159. ADMIN CONFIGURATION PRINCIPLE

The administrator should be able to configure business information without developer involvement wherever practical.

Configurable:

Services

Prices/fees where applicable

Payment methods

Gateway settings

UPI

FAQs

Blog

Pages

Projects

Locations

CTAs

Contact details

AI knowledge

Notifications

SEO

Users

Permissions

160. WHAT MUST REMAIN CODE-LEVEL

Certain functions should not be exposed as arbitrary CMS configuration:

Payment security logic

Webhook verification

Database access logic

Authentication mechanisms

Authorization logic

Sensitive backend processing

Server-side gateway signatures

AI security controls

Core business rules

System security settings

161. DEVELOPMENT PRINCIPLE

Do not build a collection of disconnected pages.

Build one integrated business platform.

The relationship must be:

WEBSITE

↓

CMS

↓

AI

↓

ENQUIRY



CRM



CUSTOMER



SITE VISIT



ESTIMATE



PROPOSAL



PAYMENT



PROJECT



MILESTONES



COMPLETION



CUSTOMER SUPPORT



REPORTING

162. FINAL MODULE MAP

The complete platform consists of:

1. Public Website
2. CMS
3. Admin Dashboard
4. CRM
5. Customer Management

6. Lead Management
7. Enquiry Management
8. Service Management
9. Construction Services
10. Renovation Services
11. Demolition Services
12. Reconstruction Services
13. Redevelopment Services
14. Property & Land Services
15. NRI Services
16. Documentation Services
17. Construction Finance Assistance
18. Project Portfolio
19. Project Management
20. Site Visit Management
21. Estimate Management
22. Proposal Management
23. Payment Gateway Management
24. UPI Payment
25. UPI QR
26. Payment Links
27. Payment Requests
28. Milestone Payments
29. Payment Receipts
30. Refund Management
31. Customer Portal
32. AI Chatbot
33. AI Knowledge Base
34. AI Lead Qualification
35. AI CRM Integration
36. Blog CMS

- 37. FAQ CMS
- 38. Testimonials
- 39. Media Library
- 40. Document Management
- 41. Notifications
- 42. Support Tickets
- 43. SEO Management
- 44. Service Area Management
- 45. Analytics
- 46. Reports
- 47. User Management
- 48. Role & Permissions
- 49. Audit Logs
- 50. System Settings
- 51. Website Settings
- 52. Security
- 53. Backup & Recovery
- 54. Marketing/Campaign Tracking
- 55. Communication Integration Readiness
- 56. Future Vendor Management Readiness
- 57. Future Construction Inventory Readiness

163. CORE BUSINESS ENTITIES

The principal database entities should include:

User

Role

Customer

Lead

Enquiry

Service

ServiceCategory

ServicePaymentConfiguration

Project

ProjectCategory

ProjectImage

ProjectMilestone

ProjectTask

SiteVisit

Estimate

EstimateItem

Proposal

ProposalItem

PaymentGateway

PaymentMethod

UPIConfiguration

PaymentRequest

PaymentTransaction

PaymentRefund

PaymentReceipt

Blog

BlogCategory

BlogTag

FAQ

Testimonial

Page

PageSection

Media

Document

Location

Notification

SupportTicket

AIConversation

AIMessage
AIKnowledge
AILeadSummary
CRMActivity
FollowUp
AuditLog
SystemSetting
SEOSetting

164. MASTER RELATIONSHIP

The platform's core relationship should be:

Customer
↓
Enquiry
↓
Lead
↓
CRM Activity
↓
Site Visit
↓
Estimate
↓
Proposal
↓
Payment Request
↓
Payment Transaction
↓
Project
↓

Project Milestones



Additional Payments



Completion



Support

This relationship must be reflected consistently throughout the application.

165. ADMINISTRATOR CONTROL

The administrator must have centralized control over:

Website

CMS

Services

Projects

CRM

AI

Payments

Blog

SEO

Users

Permissions

Reports

Notifications

System settings

The dashboard must make the system manageable without requiring technical knowledge for normal operational tasks.

166. PUBLIC CUSTOMER JOURNEY

A typical customer should be able to:

Visit website

↓

Understand services

↓

Explore projects

↓

Ask AI questions

↓

Submit enquiry

↓

Become CRM lead

↓

Receive follow-up

↓

Schedule site visit

↓

Receive estimate

↓

Receive proposal

↓

Accept proposal

↓

Receive payment request

↓

Pay through gateway/UPI

↓

Receive receipt

↓

Access project updates

↓

Make milestone payments

↓

Complete project



Provide testimonial

167. AI CUSTOMER JOURNEY

AI-assisted journey:

Visitor



AI Chat



Intent Detection



Service Recommendation



Requirement Questions



Lead Qualification



CRM Lead



AI Summary



Priority Suggestion



Human Handoff if required



Site Visit



Estimate



Proposal

↓

Payment

↓

Project

168. PAYMENT CUSTOMER JOURNEY

Service Page

↓

Payment CTA

↓

Payment Request

↓

Secure Payment Page

↓

Gateway / UPI / QR / Other

↓

Payment Verification

↓

Success / Pending / Failed

↓

Receipt

↓

CRM Update

↓

Customer Notification

169. ADMIN BUSINESS JOURNEY

Admin login

↓

Dashboard

↓

New Lead



Assign Staff



Follow-up



Site Visit



Estimate



Proposal



Payment Request



Payment



Project



Milestones



Completion



Reports

The dashboard should provide visibility throughout this lifecycle.

170. FINAL QUALITY REQUIREMENT

The final application must not look like a basic Django admin interface.

The public website must have a premium custom frontend.

The CRM and CMS must have a professional Bootstrap 5 dashboard.

The AI chatbot must have a polished conversational interface.

The payment interface must look trustworthy and secure.

The customer portal must be simple and professional.

The entire system must maintain consistent branding, typography, spacing, buttons, cards, tables, forms and navigation.

171. DEVELOPMENT QUALITY

Code must be:

Clean

Modular

Maintainable

Secure

Documented

Reusable

Scalable

Testable

Django-standard

Avoid unnecessary complexity.

Avoid duplicated code.

Use reusable templates and components.

Use proper Django forms.

Use class-based views where appropriate.

Use service-layer logic for complex operations.

Use transactions for critical financial operations.

172. DATABASE DESIGN QUALITY

Every important entity must have:

Primary key

Created timestamp

Updated timestamp

Status where appropriate

Owner/creator where appropriate

Indexes where required

Foreign keys

Proper relationships

Unique constraints

Soft-delete strategy where appropriate

173. SECURITY PRINCIPLE

Security takes priority over convenience.

Especially for:

Payments

Customer information

Documents

Admin accounts

AI

Gateway credentials

Webhook processing

Financial data

Never implement a shortcut that creates a security vulnerability.

174. PRODUCTION READINESS

Before production launch:

Configure production settings

Configure MySQL

Configure static/media storage

Configure HTTPS

Configure email

Configure payment gateway

Configure UPI

Configure webhook

Configure AI

Configure backups

Create admin users

Create roles
Create services
Create locations
Create CMS content
Create legal pages
Test enquiry forms
Test CRM
Test payments
Test receipts
Test AI
Test customer portal
Test permissions
Run security checks

175. TESTING REQUIREMENT

Test all major workflows.

Website:

Navigation

Forms

Search

Mobile responsiveness

SEO

CMS:

Create

Edit

Publish

Unpublish

Delete/archive

CRM:

Lead creation

Assignment

Follow-up
Site visit
Estimate
Proposal
Project
Payments:
Gateway
UPI
QR
Payment request
Webhook
Verification
Receipt
Refund
AI:
Conversation
Qualification
Lead creation
Summary
Payment status
Human handoff
Customer portal:
Login
Permissions
Documents
Payments
Projects
Reports:
Lead reports
Revenue
Service demand

176. FINAL ACCEPTANCE CRITERIA

The project is considered complete only when:

All primary services are implemented.

The website is fully dynamic.

CMS content can be managed without code changes.

CRM manages the full lead lifecycle.

AI chatbot works across all major services.

AI can qualify enquiries.

AI can create controlled CRM leads.

AI can hand over to staff.

Payment gateways can be configured.

UPI can be configured.

UPI QR can be configured.

Service-specific payment settings work.

CRM payment requests work.

Payment verification is secure.

Receipts are generated.

Projects can be managed.

Milestones can be tracked.

Customers can access their portal.

Blogs can be managed dynamically.

FAQs can be managed dynamically.

Projects can be managed dynamically.

SEO can be configured.

Users and permissions work.

Audit logging works.

Reports work.

Responsive design works.

Security validation passes.

The application does not depend on React, Node.js, PHP, Laravel, Vue, Angular, Tailwind or MongoDB.

177. FINAL PLATFORM VISION

This project should ultimately function as a:

CONSTRUCTION + RENOVATION + REDEVELOPMENT + PROPERTY SERVICES DIGITAL BUSINESS PLATFORM

with four major layers:

PUBLIC EXPERIENCE

Website + Services + Projects + Blog + AI



BUSINESS OPERATIONS

CMS + CRM + Customers + Leads + Site Visits + Estimates + Proposals



TRANSACTION & PROJECT OPERATIONS

Payments + Receipts + Projects + Milestones + Documents + Support



INTELLIGENCE & MANAGEMENT

AI + Analytics + Reports + Automation + Notifications + Audit + Administration

178. MASTER SYSTEM FLOW

PUBLIC WEBSITE



SERVICE DISCOVERY



AI / ENQUIRY



LEAD



CRM



QUALIFICATION

↓

SITE VISIT

↓

ESTIMATE

↓

PROPOSAL

↓

CUSTOMER APPROVAL

↓

PAYMENT

↓

PROJECT

↓

PROJECT MANAGEMENT

↓

MILESTONE PAYMENT

↓

PROGRESS UPDATES

↓

FINAL PAYMENT

↓

HANDOVER

↓

CUSTOMER SUPPORT

↓

TESTIMONIAL

↓

REPORTING & ANALYTICS

This is the final integrated business architecture.

Before writing or modifying code:

1. Inspect the entire existing project.
2. Identify all existing applications.
3. Identify existing models.
4. Identify existing URLs.
5. Identify existing templates.
6. Identify existing static files.
7. Identify existing database relationships.
8. Identify existing authentication and permissions.
9. Identify existing CMS functionality.
10. Identify existing CRM functionality.
11. Identify existing AI functionality.
12. Identify existing payment functionality.
13. Identify incomplete modules.
14. Do not duplicate existing functionality.
15. Do not delete working functionality.
16. Do not change technology stack.
17. Implement missing modules systematically.
18. Create migrations.
19. Test every integration.
20. Preserve backward compatibility.

The application must evolve as one integrated platform rather than as separate disconnected features.

180. FINAL TECHNOLOGY RESTRICTION

MANDATORY:

Python

Django

MySQL

HTML5

CSS3

Bootstrap 5

Vanilla JavaScript

Google AI Studio / Gemini

PROHIBITED:

React

Next.js

Node.js

PHP

Laravel

Vue

Angular

Tailwind

MongoDB

Any unnecessary frontend framework

Any unnecessary backend framework

181. FINAL PROJECT DEFINITION

The final product is not simply:

“A Construction Website.”

It is:

A COMPLETE DIGITAL CONSTRUCTION & PROPERTY BUSINESS MANAGEMENT PLATFORM

combining:

Website

-

CMS

-

CRM

-

AI Construction Assistant

-

Payment Gateway Platform

-

Customer Portal

-

Project Management

-

Estimate & Proposal Management

-

Blog & Knowledge Centre

-

Document Management

-

Lead & Sales Management

-

Analytics & Reporting

-

Notifications

-

SEO

-

Security & Audit

-

Administrative Control

The system must be designed so that the organization can operate its complete customer acquisition, sales, payment and project lifecycle from a single professional platform.